

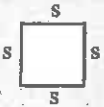
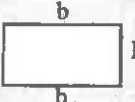
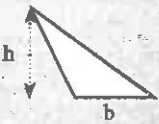
Chapter 1 Indices 1

$2 \times 2 \times 2 = 2^3$ ← Index ← Base		$(2)^2 = 2 \times 2 = 4$ $(2)^3 = 2 \times 2 \times 2 = 8$ $(2)^4 = 2 \times 2 \times 2 \times 2 = 16$	Index Law 1 $a^m \times a^n = a^{m+n}$
Index Law 2 $a^m \div a^n = a^{m-n}$	Index Law 3 $(a^m)^n = a^{m \times n}$	Zero Index $a^0 = 1$	Negative Index $a^{-m} = \frac{1}{a^m}$

Chapter 2 Algebra 1

$7a + 5 - 4a$ $= 3a + 5$	$3d \times 5e$ $= 3 \times d \times 5 \times e$ $= 15de$	$5mn \times 2m^2n$ $= 5 \times 2 \times m \times m^2 \times n \times n$ $= 10m^3n^2$	$18ay - 4a$ $= \frac{18ay}{4a}$ $= \frac{9y}{2}$
Distributive Law: $a(b + c) = ab + ac$ $3(2b - 5)$ $= 6b - 15$	$-4(x - 1) - 2(x - 4)$ $= -4x + 4 - 2x + 8$ $= -6x + 12$	Factorisation: $a(b + c) = ab + ac$ $4x - 12$ $= 4(x - 3)$	$-6x^2 + 15x$ $= -3x(2x - 5)$

Chapter 3 Area

Square  $\text{Area} = s \times s = s^2$	Rectangle  $\text{Area} = l \times b$
Triangle  $\text{Area} = \frac{1}{2}bh$	Cylinder  $\text{Surface Area} = 2\pi r^2 + 2\pi rh$

Chapter 4 Linear & Non-Linear Graphs

Linear Graphs are of the form: $y = mx + c$ a) m is the increase each step. b) c is the value of y when $x = 0$ ie $(0, c)$.	 Increase of 2 each $x \rightarrow 2x$ When $x=0, y=-3 \rightarrow y = 2x - 3$
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